Bharat Yalavarthi

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EDUCATION

University at Buffalo, The State University of New York — **Buffalo, NY**

Ph.D. in Computer Science, GPA: 3.8/4.0, UB Presidential Fellowship — *Jan 2025 - Dec 2027 (expected)*

Advisors: Dr. Nalini Ratha, Dr. Venu Govindaraju

M.S. in Computer Science, GPA: 3.9/4.0 — *Aug 2022 - May 2024*

SELECTED PUBLICATIONS

Full list on Google Scholar

- **Label-Free Mitigation of Spurious Correlations in VLMs Using Sparse Autoencoders** — ICLR 2026
  B. C. Yalavarthi*, N. Ratha, V. Govindaraju*
- **Aligning Characteristic Descriptors with Images for Human-Expert-like Explainability** — NeurIPS 2024 IAI Workshop
  B. C. Yalavarthi*, N. Ratha*
- **Enhancing Privacy in Face Analytics Using Fully Homomorphic Encryption** — IEEE FG 2024
  B. C. Yalavarthi*, A. R. Kaushik, A. Ross, V. Boddeti, N. Ratha*

RESEARCH EXPERIENCE

University at Buffalo — **Jan 2025 - Present**

PhD Student — *Buffalo, NY*

- **(ICLR 2026)** Developed DIAL, a label-free method using Sparse Autoencoders (SAEs) that removes spurious correlations from VLM representations. Improved worst-group accuracy by up to 37 points across 5 standard benchmarks without labels, extra data, or retraining [Paper]
- **Generation Failures:** Predicted diffusion and flow matching failures from intermediate states early in generation (under review).
- **Prompt Injection Interpretability:** Used probes and ablations on GPT-OSS-20B to study injection attacks. The model detects the instruction conflict, yet this signal does not predict whether the attack succeeds.
- **Causal study of LLM knowledge formation:** Showed that contextual diversity, not repetition, drives factual recall and linear accessibility. 2K diverse exposures outperform 8K duplicates (under review).
- **SAE Feature Granularity:** Across Gemma Scope SAE widths, wide SAEs form compositional feature families rather than atomic features, while narrow SAEs collapse frequent concepts into coarse superpositions.

SUNY Research Foundation — **Aug 2024 - Jan 2025**

Researcher — *Buffalo, NY*

- **Benchmark Creation:** Designed a Gestalt-grounded benchmark (1,200+ stimuli, 4 perceptual principles) revealing systematic visual reasoning failures in GPT and Claude models that standard VQA benchmarks miss.
- **Reliability Evaluation:** Stress-tested state-of-the-art multimodal LLMs on OCR with targeted perturbations, mapping where vision encoders break down and model outputs become unreliable.

University at Buffalo — **Jan 2023 - May 2024**

Research Assistant — *Buffalo, NY*

- **Explainability:** Used Mistral-7B and CLIP to generate human-expert-like justifications for face recognition and medical predictions, improving explanation faithfulness through characteristic descriptors (NeurIPS 2024 IAI Workshop).
- **Privacy & Secure ML:** Exposed attribute leakage in face-template protection schemes and built an FHE-based mitigation that cut leakage by 35% over prior methods with no accuracy loss (IEEE FG 2024, 2025).

INDUSTRY EXPERIENCE

Harman International (Samsung) — **Oct 2020 - Jul 2022**

Software Engineer, ADAS & Navigation — *Bangalore, India*

- Delivered 20+ ADAS and route guidance features for VW Trucks and Coaches, with a modular redesign that reduced software maintenance costs by 25%.
- Led the product line expansion of ADAS modules and integrated an onboard traffic sign recognition module, cutting incorrect sign events by 90%.

SKILLS & ACADEMIC SERVICE

- **Skills:** Interpretability (SAEs, activation patching, causal tracing, probing, steering, transcoders, CBMs), evaluation and benchmark design, PyTorch, Hugging Face, SAELens, multi-GPU training (DDP), HPC clusters (SLURM), Docker, LLMs, VLMs, Diffusion Models, Python, C++
- **Reviewer:** NeurIPS, ICLR, CVPR, WACV. Presented at NSF CITeR Program Review 2024.